LEVEL 3 · REMOTE ACCESS (OUTSIDE THE PLANT)

Remote support (vendor / O&M)
VPN with multi-factor login; off by default, enabled per session by an Admin and logged

VPN, off by default

FW-01 · Firewall with VPN
Fortinet FortiGate Rugged 60F; deny by default; only conduit to the outside

No plant data leaves the site in the proposed design (report section 9).

LEVEL 2 · SUPERVISORY – SCADA ZONE (CONTROL ROOM)

SRV-A
SICAM SCC server, active
SIMATIC IPC647E

SRV-B
SICAM SCC server, hot standby
SIMATIC IPC647E

OWS-01
Operator workstation, two monitors

ENG-01
Engineering laptop, SICAM Device Manager

GPS-01
Meinberg LANTIME M300, NTP time

UPS-01
APC Smart-UPS SRT 3000 VA, SNMP

ALM-01
Alarm sounder and beacon, hardwired to row 1

SW-CORE-A · Moxa EDS-G512E
Managed core switch A; VLANs, port security, SNMP v3; fibre ring side A

SW-CORE-B · Moxa EDS-G512E
Managed core switch B; second path for every dual-connected device; ring side B

LEVEL 1 · CONTROL – SICAM A8000 HOT STANDBY PAIR (11 KV SWITCHROOM, SCADA CABINET)

CP-8050 A · active
CP-8050 processor 6MF2805-0AA00
2 × PS-8642 power supply
CI-8520 Ethernet, one port per zone (VLAN 10/20/30/40)
IEC 60870-5-104 server to SRV-A / SRV-B
IEC 61850 / Modbus TCP client to the field devices
Interlocks, secured breaker commands, 1 ms time stamps

HSBY

CP-8050 B · hot standby
CP-8050 processor 6MF2805-0AA00
2 × PS-8642 power supply
CI-8520 Ethernet, one port per zone (VLAN 10/20/30/40)
Follows A; takes over if A fails
Switchover time: Siemens to confirm
Same configuration and flash card as A

SICAM I/O rows (hardwired)

Row	Location	Interface	Modules	Sig.
1	11 kV switchroom	CI-8532, copper	DI-8112 ×5, DO-8221 ×2, DO-8212	87
2	SCADA cabinet	CI-8532, copper	DI-8112 ×2, AI-8320	25
3	President station	CI-8533, fibre pair	DI-8112 ×3, DO-8221	43
4	PM station	CI-8533, fibre pair	DI-8112 ×3, DO-8221	43
5	BESS / inverter yard	CI-8533, fibre pair	DI-8112 ×2	26
6	Generator area	CI-8533, fibre pair	DI-8112 ×2, AI-8320	20
	Total		24 modules, 27 % spare	244

SICAM I/O link: both controllers reach every row; only the active one reads inputs and drives outputs (Siemens to confirm).

Rows 1–2 on copper in the cabinet; rows 3–6 on dedicated fibre pairs, separate from the network ring (sheet 002).

LEVEL 0 · FIELD DEVICES BY LOCATION

Fibre ring through the field and station switches (sheet 002)

Core A / B · SW-BESS · SW-GEN · SW-MV · SW-PRS · SW-PMS

11 kV SWITCHROOM

Relays AH1 … AH7
IEC 61850 MMS or Modbus TCP, both core switches
Measurements, trips, events

Panels AH1 … AH8 hardwired
Positions, trips, earth switch, L/R, spring, gas → row 1
Open / close AH1–AH7 ← DO-8221

NECR AH8
Earthing TX + NER 63.5 Ω; breaker, earth fault → row 1

Aux TX, LV board, UPS, DC
Fuse, TX, UPS, DC alarms → row 2
PM5560-AUX, UPS: Modbus TCP / SNMP

E-stop and doors
Hardwired → row 2; display only

BESS CONTAINER

EMS (embedded)
Modbus TCP, control zone
Dispatches PV, battery and generators

ST3340UX battery + PCS
3,340 kWh; read through the EMS only

Fire panel
Pre-alarm, fire, release, fault → row 5
Trips the battery by its own circuit

BESS ACB, E-stop, door
Position, trip, E-stop, door → row 5

GENERATOR AREA

DG1 controller · 550 kVA
Modbus TCP, or RTU through a gateway
Start / stop in manual mode

DG2 controller · 660 kVA
As DG1

Generators hardwired
GCB, running, alarm, E-stop → row 6
Fuel level DG1, DG2, 4–20 mA → row 6

Step-up TX, LV ACB
TX alarms, ACB position and trip → row 6

EMS start contacts
Hardwired EMS → generator controllers, outside the SCADA

PV INVERTER STATION

PLC box (Sungrow)
Modbus TCP; inverter data, power limit

Inverters NB01 … NB04
SG350HX 350 kW; power line communication to the PLC box

Weather logger blue'Log XM
Weather station only; Modbus TCP

PV ACB, MCCBs, inverter TX
Position, trips, TX alarms → row 5

PRESIDENT STATION

RMU relay ST260E
Modbus RTU through MGate MB3170

PM5560-PRS
LVDB meter, Modbus TCP

RMU, TX, LVDB hardwired
Switches, TX alarms, 10 ways → row 3
Commands ← DO-8221

RMU UPS 2 kVA
On battery, fault → row 3

PM STATION

RMU relay ST260E
Modbus RTU through MGate MB3170

PM5560-PMS
LVDB meter, Modbus TCP

RMU, TX, LVDB hardwired
Switches, TX alarms, 10 ways → row 4
Commands ← DO-8221

RMU UPS 2 kVA
On battery, fault → row 4

LEGEND

- Ethernet copper – IEC 104 / Modbus TCP / IEC 61850
- Ethernet single-mode fibre (ring)
- Link to core switch A
- Link to core switch B
- SICAM I/O link (copper or fibre pair)
- Hot standby link CP-8050 A ↔ B
- RS485 – Modbus RTU
- Hardwired I/O (DI / DO / AI)
- Power line communication (inverters)
- VPN, off by default
- Device; stripe = link type to the SCADA
- Zone or location boundary

NOTES

1. Architecture as SCADA Design Report Rev 1, sections 4 to 8, and Device and I/O List Rev 1. Device names and tags are used unchanged on sheets 002 to 005, the HMI and the I/O list.
2. The embedded EMS dispatches PV, battery and generators. The CP-8050 pair monitors, executes breaker commands with interlocks and passes Admin settings to the EMS; it does not dispatch.
3. CP-8050 A is active and B hot standby. Siemens to confirm the redundancy function and licence, the switchover time and the I/O row connection to both controllers.
4. Protection trips stay in the relays. Emergency stop and fire trips are hardwired and act directly on the equipment; the SCADA only shows them.
5. Zones are separated by VLANs, separate CI-8520 ports and firewall rules. Only FW-01 connects to the outside, with VPN off by default.
6. Quantities and models: report section 13 and the bill of material (Attachment D).

REV	DESCRIPTION	DATE
A	Issued for client approval – SICAM A8000 design	16-OCT-2026
B	Re-issued with Design Report Rev 1; SICAM A8000 confirmed	03-NOV-2026

CLIENT:
MINISTRY OF ENERGY & WATER RESOURCES – SOMALIA
SITE: VILLA SOMALIA

CONTRACTOR:
BLUEFLAG ENERGY CO. LTD

PROJECT TITLE:
SOLARIZATION VILLA SOMALIA – SCADA

SHEET TITLE:
SCADA ARCHITECTURE
System overview by level and location

SCALE: NTS	DATE: 03-NOV-2026	SIZE: A3
DESIGNED	E. M.	SIGN
DRAWN		SIGN
CHECKED		SIGN
APPROVED		SIGN

DRW NO. **BFE/PR/26/01-SCADA-001** **REV B**

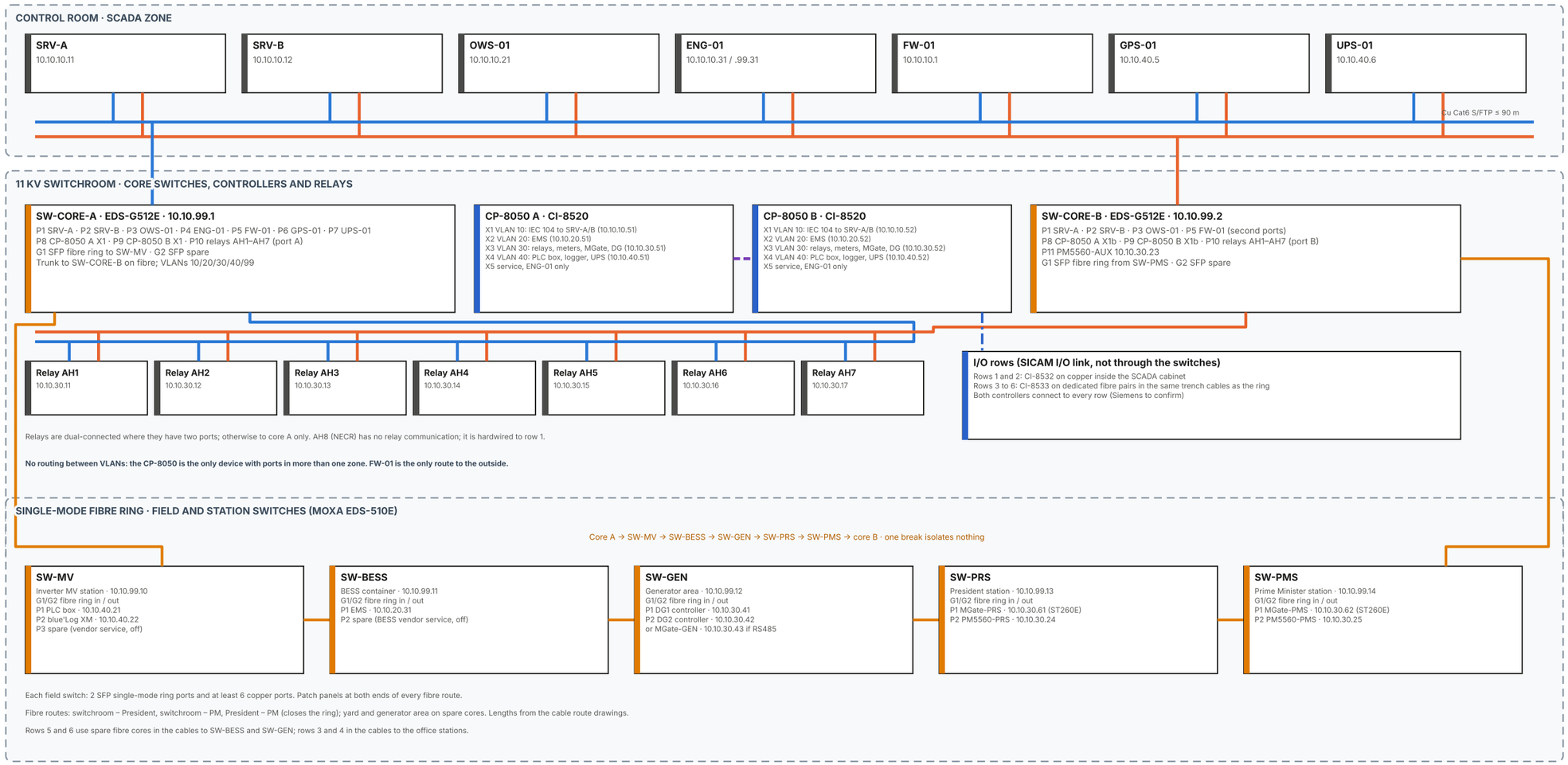


TABLE A – VLAN AND IP PLAN (PROPOSED)

VLAN	Subnet	Zone and devices	Routing
10	10.10.10.0/24	SCADA: SRV-A/B, OWS-01, ENG-01, FW-01 inside, CP-8050 X1	FW-01 to outside only
20	10.10.20.0/24	Control: CP-8050 X2, EMS	None
30	10.10.30.0/24	Substation: relays AH1–AH7, PM5560 ×3, MGate ×2, DG controllers, CP-8050 X3	None
40	10.10.40.0/24	Monitoring: PLC box, blue'Log XM, UPS, GPS-01, CP-8050 X4	None; read only
99	10.10.99.0/24	Switch management: core and field switches	ENG-01 only

TABLE B – RS485 MODBUS RTU BUSES

Bus	Devices (Modbus ID)	Settings	Remark
MGate-PRS port 1	ST260E President (1)	9600 8N1 (to confirm)	Shielded twisted pair, 120 Ω at both ends
MGate-PMS port 1	ST260E Prime Minister (1)	9600 8N1 (to confirm)	As above
MGate-GEN (only if needed)	DG1 (1), DG2 (2)	19200 8N1 (to confirm)	Only if the generator controllers are RS485 only

TABLE C – CABLE TYPES

Type	Specification
Ethernet copper	Cat6 S/FTP industrial, shielded RJ45, ≤ 90 m
Network fibre	Single-mode OS2, 12-core armoured, LC duplex
I/O row fibre	CI-8533 optical SICAM I/O link; fibre type and maximum length to confirm with Siemens
RS485	2-pair shielded twisted pair 120 Ω, shield earthed at the master end
Hardwired I/O	Multicore 1.0 mm² screened, 110 V DC, separate from power cables

LEGEND

- Ethernet copper – IEC 104 / Modbus TCP / IEC 61850
- Ethernet single-mode fibre (ring)
- Link to core switch A
- Link to core switch B
- SICAM I/O link (copper or fibre pair)
- Hot standby link CP-8050 A ↔ B
- RS485 – Modbus RTU
- Device; stripe = link type to the SCADA
- Zone or location boundary

NOTES

1. Read with sheet 001. Addresses and port numbers are proposals for approval; final values are fixed in the IP plan at the factory test.
2. All switches managed: unused ports disabled, port security, SNMP v3 and syslog to SRV-A/B, HTTPS and SSH management on VLAN 99.
3. No routing between VLANs. The CP-8050 is the only device with ports in more than one zone (CI-8520), and FW-01 is the only route to the outside.
4. Ring protocol and recovery time are set at configuration (Moxa Turbo Ring or RSTP).
5. I/O rows 3 to 6 run on dedicated fibre pairs (CI-8533), separate from the ring. Siemens to confirm whether they can be laid as a ring.
6. All devices take time from GPS-01 by NTP. Relays and the CP-8050 time-stamp events at 1 ms.
7. The communication schematic of BFE/PR/26/01 (sheet 15) is replaced by this sheet.

REV	DESCRIPTION	DATE
A	Issued for client approval – SICAM A8000 design	16-OCT-2026
B	Re-issued with Design Report Rev 1; SICAM A8000 confirmed	03-NOV-2026

CLIENT:
MINISTRY OF ENERGY & WATER RESOURCES – SOMALIA
SITE: VILLA SOMALIA

CONTRACTOR:
BLUEFLAG ENERGY CO. LTD

PROJECT TITLE:
SOLARIZATION VILLA SOMALIA – SCADA

SHEET TITLE:
COMMUNICATION NETWORK
Topology, ports, IP plan and serial buses

SCALE: NTS	DATE: 03-NOV-2026	SIZE: A3
DESIGNED	E. M.	SIGN
DRAWN		SIGN
CHECKED		SIGN
APPROVED		SIGN

DRW NO. **BFE/PR/26/01-SCADA-002** **REV B**

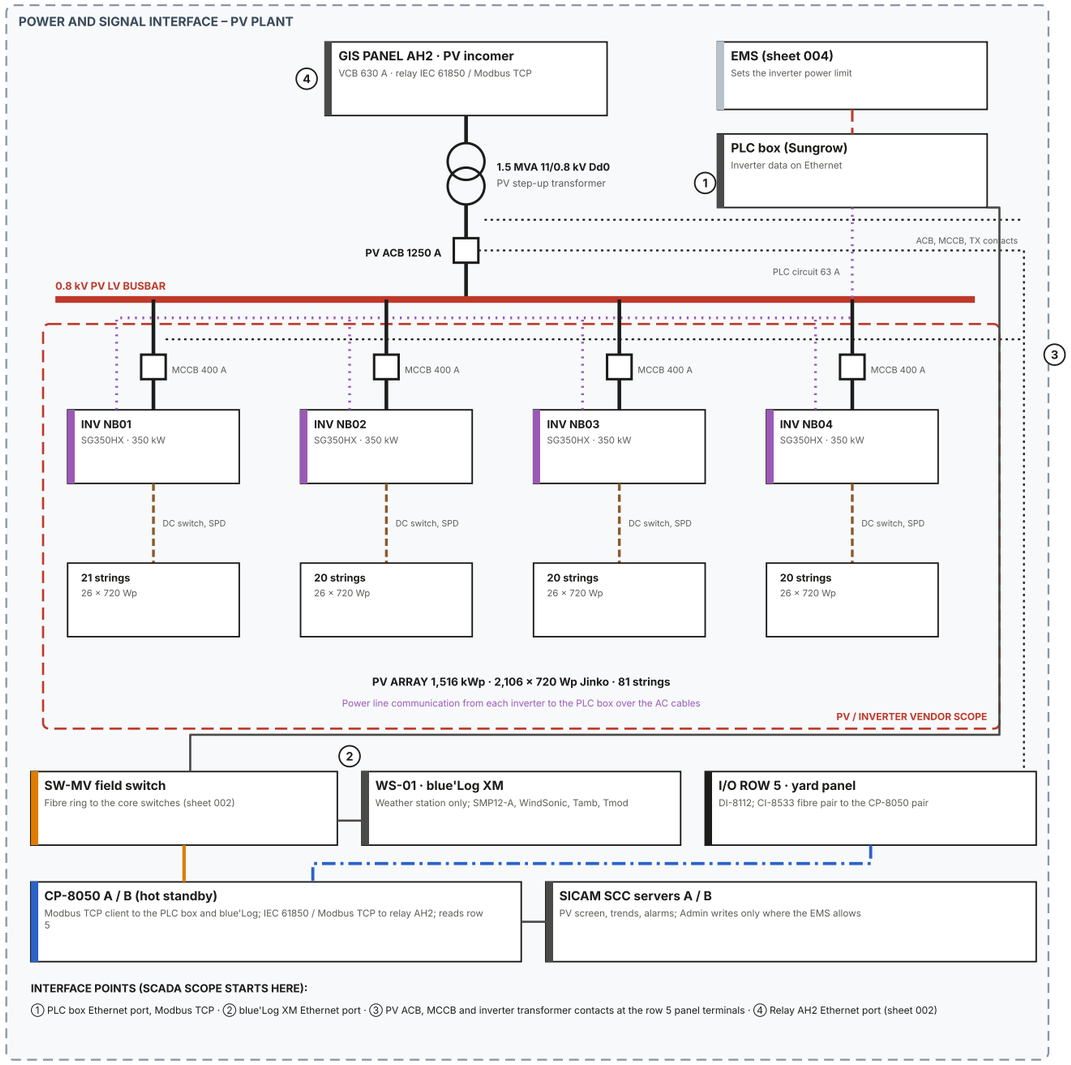


TABLE A – SIGNAL EXCHANGE (PER INVERTER UNLESS STATED)

Dir.	Signal (I/O list tags)	Type	Source / path	Update
IN	P, Q, U and I L1–L3, f, PF (NBxx-P … -PF)	AI com	PLC box, Modbus TCP	2 s
IN	DC U, I, P (NBxx-U-DC, -I-DC, -P-DC)	AI com	PLC box	5 s
IN	Energy today and total (NBxx-E-DAY, -E-TOT)	AI com	PLC box	60 s
IN	Internal temperature, insulation (NBxx-T-INT, -R-ISO)	AI com	PLC box	10 s
IN	State, fault and alarm codes (NBxx-STATE, -FLT-CODE, -ALM-CODE)	ST com	PLC box	2 s
IN	Inverter link healthy (NBxx-COMM-OK, PLCBOX-NBxx-COMM)	DI com	PLC box	5 s
OUT	Active power limit (NBxx-P-LIM-SP)	AO com	Admin, only if the EMS allows	on change
OUT	Start / stop (NBxx-START-STOP-CMD)	DO com	Admin, only if the EMS allows	command
IN	Irradiance, wind, ambient and module temperature (WS-*)	AI com	blue'Log XM	5 s
IN	PV ACB closed / tripped (PV-ACB-CLSD, -TRIP)	DI	Row 5	event, 1 ms
IN	Inverter AC breaker tripped (NB01 … NB04-MCCB-TRIP)	DI	Row 5	event, 1 ms
IN	Inverter transformer alarms and trips (T-INV-*)	DI	Row 5	event, 1 ms
IN	AH2 measurements, trips, events (AH2-*)	AI/DI com	Relay AH2	1 s / event

Signal count (I/O list Rev 1): 12 hardwired DI · 94 read and 8 write communication points (inverters, PLC box, weather station) · plus relay AH2.

TABLE B – CONTROL FUNCTIONS AND BEHAVIOUR

Function	Description
Curtailment	The EMS limits the inverters through the PLC box when the battery is full or a generator would fall below minimum load; ramped, not tripped. The SCADA shows it (EMS-PV-CURT).
Writes from the SCADA	Power limit and start / stop: Admin role only, and only if the EMS supplier allows remote writes; otherwise display only.
Close permissive AH2	The 11 kV busbar must be live, formed by the battery or a generator; PV cannot energise the busbar.
Loss of the grid former	Inverters trip on their own voltage and frequency protection (anti-islanding).
PLC box link lost	Level 2 alarm; values shown stale (###); commands to the inverters blocked.
Performance ratio	Calculated in the SCADA from PV energy and plane-of-array irradiance.

TABLE C – INTERFACE RESPONSIBILITY

Item	PV / inverter vendor	SCADA contractor
PLC box Modbus register map	Supply	Implement
Settings that may be written remotely	Define and confirm in writing	Implement, Admin role only
PLC box Ethernet port and address	Provide the port	Issue the address; cable to SW-MV
MCCB, ACB and TX contacts	Wire to a terminal strip	Wire to the row 5 panel
Weather station sensors and logger	Supply, install	Integrate over Modbus TCP
Default passwords	Change and hand over	Record in the site register
Interface test	Attend	Lead and record

LEGEND

- Power circuit
- DC circuit
- Power line communication (inverters)
- Ethernet copper – IEC 104 / Modbus TCP / IEC 61850
- Ethernet single-mode fibre (ring)
- SICAM I/O link (copper or fibre pair)
- Hardwired I/O (DI / DO / AI)
- Vendor scope boundary
- Device; stripe = link type to the SCADA

NOTES

1. Ratings from the overall SLD and inverter wiring diagram of BFE/PR/26/01. NB01 is taken as the inverter with 21 strings; to be confirmed by the PV designer.
2. Inverters communicate by power line communication over the AC cables to the PLC box, then Ethernet to SW-MV. The blue'Log XM serves the weather station only.
3. PV dispatch and curtailment stay with the EMS. The SCADA writes only settings the EMS supplier permits, Admin role only.
4. Update rates are proposals, to be fixed with the PLC box register map.
5. Tags as in the I/O list Rev 1 (NB01 … NB04, PLCBOX-, PV-, T-INV-, WS-).

REV	DESCRIPTION	DATE
A	Issued for client approval – SICAM A8000 design	16-OCT-2026
B	Re-issued with Design Report Rev 1; SICAM A8000 confirmed	03-NOV-2026

CLIENT:
MINISTRY OF ENERGY & WATER RESOURCES – SOMALIA
SITE: VILLA SOMALIA

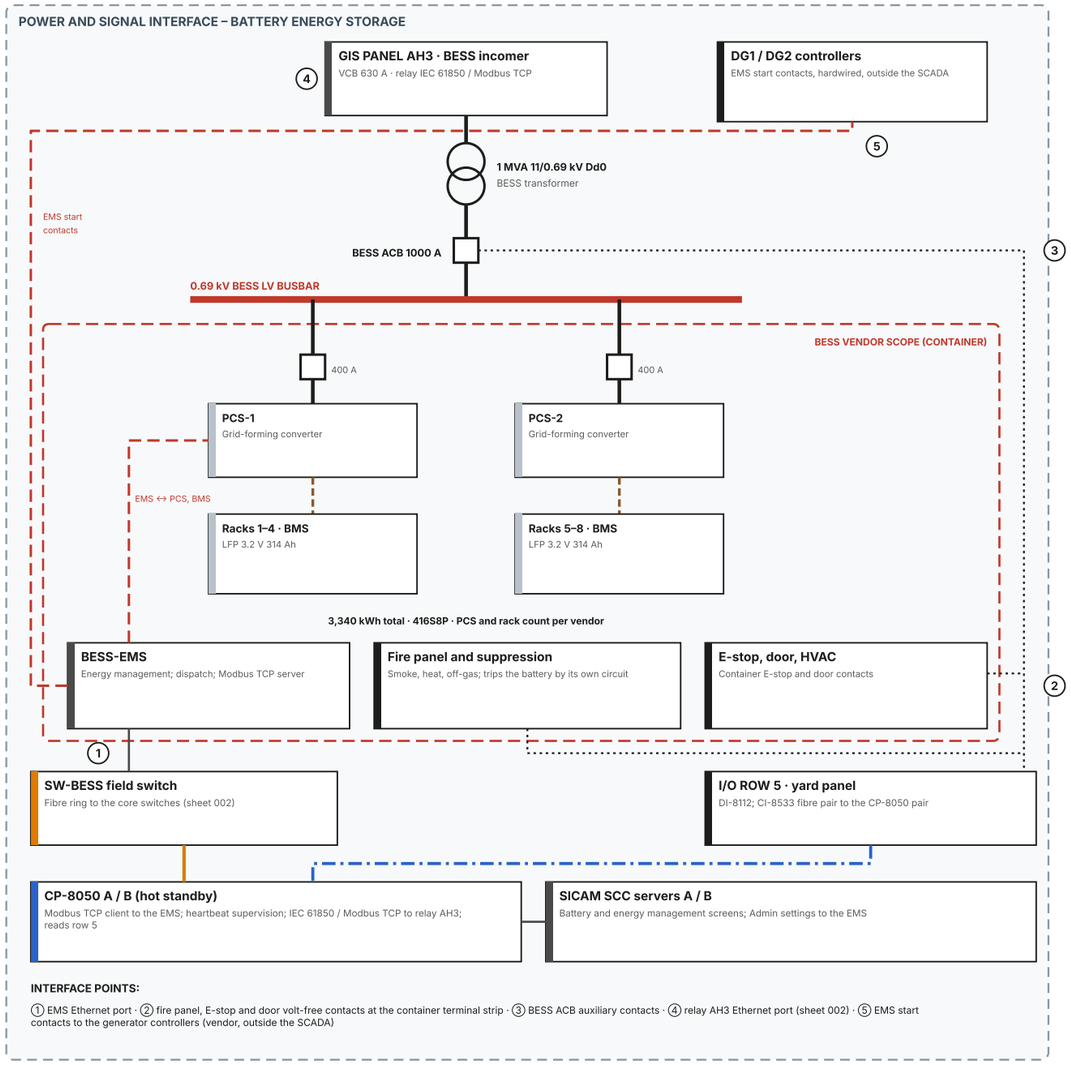
CONTRACTOR:
BLUEFLAG ENERGY CO. LTD

PROJECT TITLE:
SOLARIZATION VILLA SOMALIA – SCADA

SHEET TITLE:
PV INTERFACE
Signal exchange, control and scope boundary

SCALE: NTS	DATE: 03-NOV-2026	SIZE: A3
DESIGNED	E. M.	SIGN
DRAWN		SIGN
CHECKED		SIGN
APPROVED		SIGN

DRW NO. **BFE/PR/26/01-SCADA-003** **REV B**



INTERFACE POINTS:

① EMS Ethernet port · ② fire panel, E-stop and door volt-free contacts at the container terminal strip · ③ BESS ACB auxiliary contacts · ④ relay AH3 Ethernet port (sheet 002) · ⑤ EMS start contacts to the generator controllers (vendor, outside the SCADA)

TABLE A – SIGNAL EXCHANGE SCADA ↔ BESS

Dir.	Signal (I/O list tags)	Type	Path	Update
IN	SOC, SOH, energy charged and discharged (BESS-SOC, -SOH, -E-CHG, -E-DIS)	AI com	EMS	5 s
IN	Charge and discharge power limits (BESS-P-CHG-LIM, -P-DIS-LIM)	AI com	EMS	1 s
IN	P, Q, DC U and I (BESS-P, -Q, -U-DC, -I-DC)	AI com	EMS	1 s
IN	Cell voltage and temperature max / min (BESS-U-CELL-*, -T-CELL-*)	AI com	EMS	5 s
IN	PCS, BMS and cooling state, racks online, DC contactors	ST/DI com	EMS	2 s
IN	Alarm levels 1–3, container fire, combiner panels (BESS-ALM-*, -FIRE, -BCP*)	DI com	EMS	1 s
IN	Power balance, bus U and f, mode, state, generator request, curtailment (EMS-*)	AI/DI com	EMS	1 s
IN/OUT	Heartbeat (EMS-HEARTBEAT)	AI com	EMS ↔ CP-8050	1 s
OUT	Mode, SOC min / max, generator start / stop thresholds, power limit (EMS-*-SP, EMS-MODE-CMD)	AO com	Admin	on change
OUT	C-rate limit (Admin); fault reset level 3 (Operator), levels 1–2 (Admin)	DO com	Operator / Admin	command
IN	BESS ACB closed / tripped (BESS-ACB-CLSD, -TRIP)	DI	Row 5	event, 1 ms
IN	Fire panel: pre-alarm, fire, release, fault, agent low, isolated, manual release, supply (FIRE-*)	DI	Row 5	event, 1 ms
IN	E-stop BESS area, container door (ESTOP-BESS, DOOR-BESS)	DI	Row 5	event, 1 ms
IN	AH3 measurements, trips, events (AH3-*)	AI/DI com	Relay AH3	1 s / event

Signal count (I/O list Rev 1): 12 hardwired DI · 41 read and 9 write communication points (battery and EMS) · plus relay AH3.

TABLE B – OPERATING MODES (EMS)

Mode	Battery	PV	Generators
Day, PV surplus	Forms the grid, charging	Full; curtailed when the battery is full	Off
Day, PV shortfall / night	Forms the grid, discharging	Full / off	Off
Low battery (SOC < 20 %)	Charging from the generator	Full if available	One running; second above 80 % load
Generator only	Off	Limited to keep the generator above minimum load	Running, forming the grid
Manual	As commanded	As commanded	As commanded

TABLE C – FAILURE BEHAVIOUR

Event	Response
EMS heartbeat lost	Level 1 alarm; generators start on the dead busbar and supply the plant; battery off until the EMS is restored.
Battery fault	EMS starts a generator and moves to generator-only mode.
Fire in the container	Hardwired trip of the battery; suppression releases; EMS starts a generator; no remote reset.
E-stop pressed	Hardwired stop of battery, PV and generators; reset locally.
EMS link lost	Level 2 alarm; values shown stale; writes blocked.
AH3 protection trip	Relay opens AH3; remote reclose blocked until reset at the relay.

TABLE D – INTERFACE RESPONSIBILITY

Item	BESS vendor	SCADA contractor
EMS Modbus point table and heartbeat	Supply	Implement; supervise heartbeat
Settings that may be written remotely	Define and confirm in writing	Implement, Admin role only
Network port and address	Provide the port; agree the address	Issue the address; cable to SW-BESS
Fire, E-stop and door contacts	Volt-free contacts at a terminal strip	Wire to the row 5 panel
Alarm list and levels	Provide the alarm list	Map to levels 1 to 3
Fire and suppression system	Supply, commission	Monitor only; no remote reset
Interface test	Attend with the EMS in test mode	Lead and record

LEGEND

- Power circuit
- DC circuit
- Ethernet copper – IEC 104 / Modbus TCP / IEC 61850
- Ethernet single-mode fibre (ring)
- SICAM I/O link (copper or fibre pair)
- Hardwired I/O (DI / DO / AI)
- Vendor scope boundary
- Device; stripe = link type to the SCADA

NOTES

1. Ratings from the overall SLD of BFE/PR/26/01: 3.34 MWh LFP 416S8P, 1 MVA 11/0.69 kV, 2 × 400 A to the PCS. PCS and rack count, rating and the EMS make to be confirmed by the vendor.
2. The EMS dispatches; thresholds (start below 20 %, stop at 80 %, second generator above 80 % load) are proposed starting values, set with the EMS supplier.
3. The SCADA never releases or resets the fire suppression; it monitors only. Hardwired trips act without the controllers or communication.
4. Update rates are proposals, to be fixed with the EMS point table.
5. Tags as in the I/O list Rev 1 (BESS-, EMS-, FIRE-, ESTOP-BESS, DOOR-BESS).

REV	DESCRIPTION	DATE
A	Issued for client approval – SICAM A8000 design	16-OCT-2026
B	Re-issued with Design Report Rev 1; SICAM A8000 confirmed	03-NOV-2026

CLIENT:
MINISTRY OF ENERGY & WATER RESOURCES – SOMALIA
SITE: VILLA SOMALIA

CONTRACTOR:
BLUEFLAG ENERGY CO. LTD

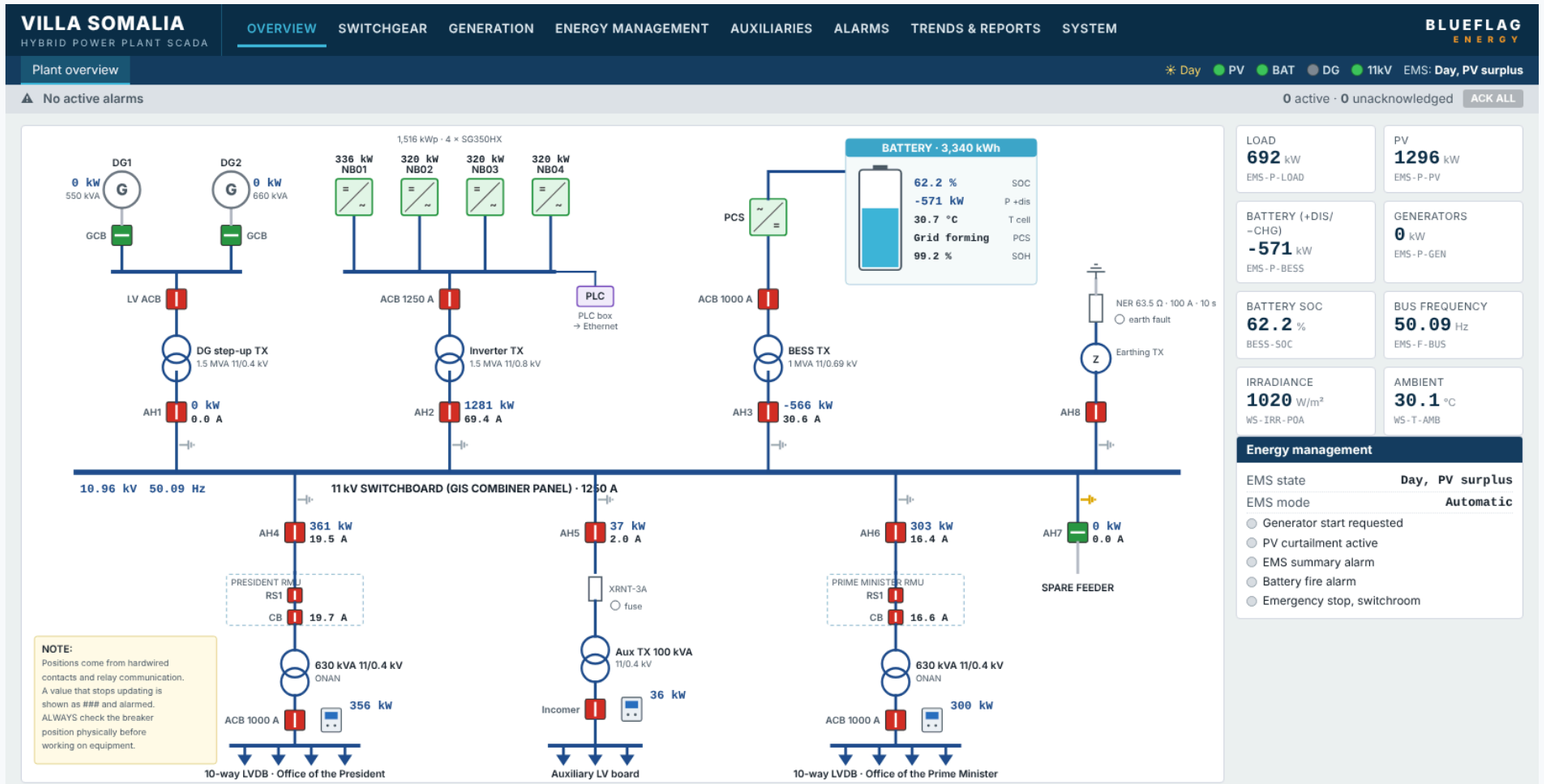
PROJECT TITLE:
SOLARIZATION VILLA SOMALIA – SCADA

SHEET TITLE:
BESS INTERFACE
Signal exchange, modes and scope boundary

SCALE: NTS	DATE: 03-NOV-2026	SIZE: A3
DESIGNED	E. M.	SIGN
DRAWN		SIGN
CHECKED		SIGN
APPROVED		SIGN

DRW NO. **BFE/PR/26/01-SCADA-004** **REV B**

SCADA OVERVIEW SCREEN – FOLLOWS THE OVERALL SLD OF BFE/PR/26/01 (VALUES ILLUSTRATIVE)



SCREEN HIERARCHY

Level	Screens
1 Plant	Overview (this sheet)
2 Area	11 kV switchboard, President and PM stations, PV, battery, generators, energy management, auxiliaries
3 Function	Alarms, event log, trends, reports, communication status, I/O list, users
4 Pop-ups	Breaker faceplate, setpoint entry, meter details

LEGEND

- Breaker closed
- Breaker open
- Breaker tripped
- Energised conductor
- De-energised conductor

NOTES

1. Screen captured from the HMI demonstration (hmi/index.html). Values are simulated and for illustration only.
2. Layout follows the overall SLD and 11 kV switchgear configuration of BFE/PR/26/01.
3. Tags follow the Device and I/O List Rev 1.
4. Breaker status comes from hardwired auxiliary contacts and is confirmed by the relay. Operators verify the position locally before working on equipment.
5. The final screens are built in SICAM SCC to the same layout and content.

TABLE A – SYMBOL TO SIGNAL REFERENCE

Symbol	Status and measurement tags	Source
AH1 … AH8 breaker	AHn-CB-OPEN, -CB-CLSD, -CB-TRIP	Row 1 DI, hardwired
AH1 … AH7 values	AHn-P, -I-L1 … -L3, -U-L1, -E-IMP	Relay AHn, IEC 61850 / Modbus TCP
AH1 … AH7 command	AHn-CB-OPEN-CMD, -CB-CLOSE-CMD	Row 1 DO-8221, select and confirm
Earth switch	AHn-ES-CLSD	Row 1 DI
DG1 / DG2	DGn-P, DGn-RUN, DGn-GCB-CLSD, DGn-MODE	Generator controller; row 6
DG step-up LV ACB	GEN-ACB-CLSD, -TRIP	Row 6 DI
PV ACB and inverters	PV-ACB-CLSD; NBxx-P, NBxx-STATE	Row 5 DI; PLC box
BESS ACB and battery	BESS-ACB-CLSD; BESS-SOC, -P, -PCS-STATE	Row 5 DI; EMS
RMU and LVDB	PRS/PMS-RS1-, -RS2-, -CB-CLSD, -LV-INC-CLSD, -LV-P	Rows 3 and 4; PM5560
Auxiliary supply	T-AUX-FUSE, AUXLV-INC-CLSD, AUXLV-P	Row 2; PM5560-AUX
NECR AH8	AH8-CB-CLSD, AH8-NER-EF	Row 1 DI
Busbar kV and Hz	EMS-U-BUS, EMS-F-BUS	EMS

TABLE B – SYMBOL AND COLOUR CONVENTIONS

Element	Convention
Breaker closed	Red filled symbol, "CLOSED" (as the screens already used on site)
Breaker open	Green filled symbol, "OPEN"
Breaker tripped	Amber, "TRIPPED", level 1 alarm
Energised / de-energised	Dark blue / grey conductor
Stale value	### in magenta, never shown as live
Alarm levels	1 red · 2 orange · 3 yellow; banner shows the newest unacknowledged alarm

TABLE C – OPERATOR INTERACTION

Action	Result
Click a breaker	Faceplate: status, measurements, live interlocks, select-before-operate
Command	Select, then confirm within 30 s; the CP-8050 checks interlocks and local/remote
Roles	Operator: monitor, acknowledge, switch, generator start / stop in manual, reset level 3. Admin: also EMS settings and modes, users, VPN
Logging	Every command, setting change and login stored with user and time

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CLIENT:
MINISTRY OF ENERGY & WATER RESOURCES – SOMALIA
SITE: VILLA SOMALIA

CONTRACTOR:
BLUEFLAG ENERGY CO. LTD

PROJECT TITLE:
SOLARIZATION VILLA SOMALIA – SCADA

SHEET TITLE:
SCADA OVERVIEW SLD
HMI overview screen and signal reference

SCALE: NTS	DATE: 03-NOV-2026	SIZE: A3
DESIGNED	E. M.	SIGN
DRAWN		SIGN
CHECKED		SIGN
APPROVED		SIGN

DRW NO. **BFE/PR/26/01-SCADA-005** **REV B**